

Week 1: make_moons active-learning sanity check

We reveal six labels, fit a GP regressor to labels in $\{-1, +1\}$, and repeatedly query the pool point nearest the believed boundary: $\operatorname{argmin} |\mu(x)|$. The initial design contains three random points from each class so that the first GP sees both regimes.

```
In [4]: from pathlib import Path
import sys

ROOT = Path.cwd().resolve()
if ROOT.name == 'notebooks':
    ROOT = ROOT.parent
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))

from src.make_moons_sanity_check import run_experiment, save_results
```

```
In [5]: result = run_experiment(seed=0, initial_size=6, total_budget=50)
save_results(result, ROOT / 'outputs' / 'make_moons')
{n: round(float(s['uncertain_fraction']), 4) for n, s in result.snapshots.items()}
```

```
Out[5]: {6: 0.8208, 20: 0.5639, 50: 0.3092}
```

The credible-region area is the fraction of the unit square satisfying $|\mu(x)| \leq 1.96 \sigma(x)$. It is a simple, inspectable proxy for posterior uncertainty around the level set. The colour display is derived from the GP latent mean for visualization; it is not a calibrated class probability.

```
In [6]: from IPython.display import Image, display
display(Image(filename=str(ROOT / 'outputs' / 'make_moons' / 'boundary_contraction.png')))
```

